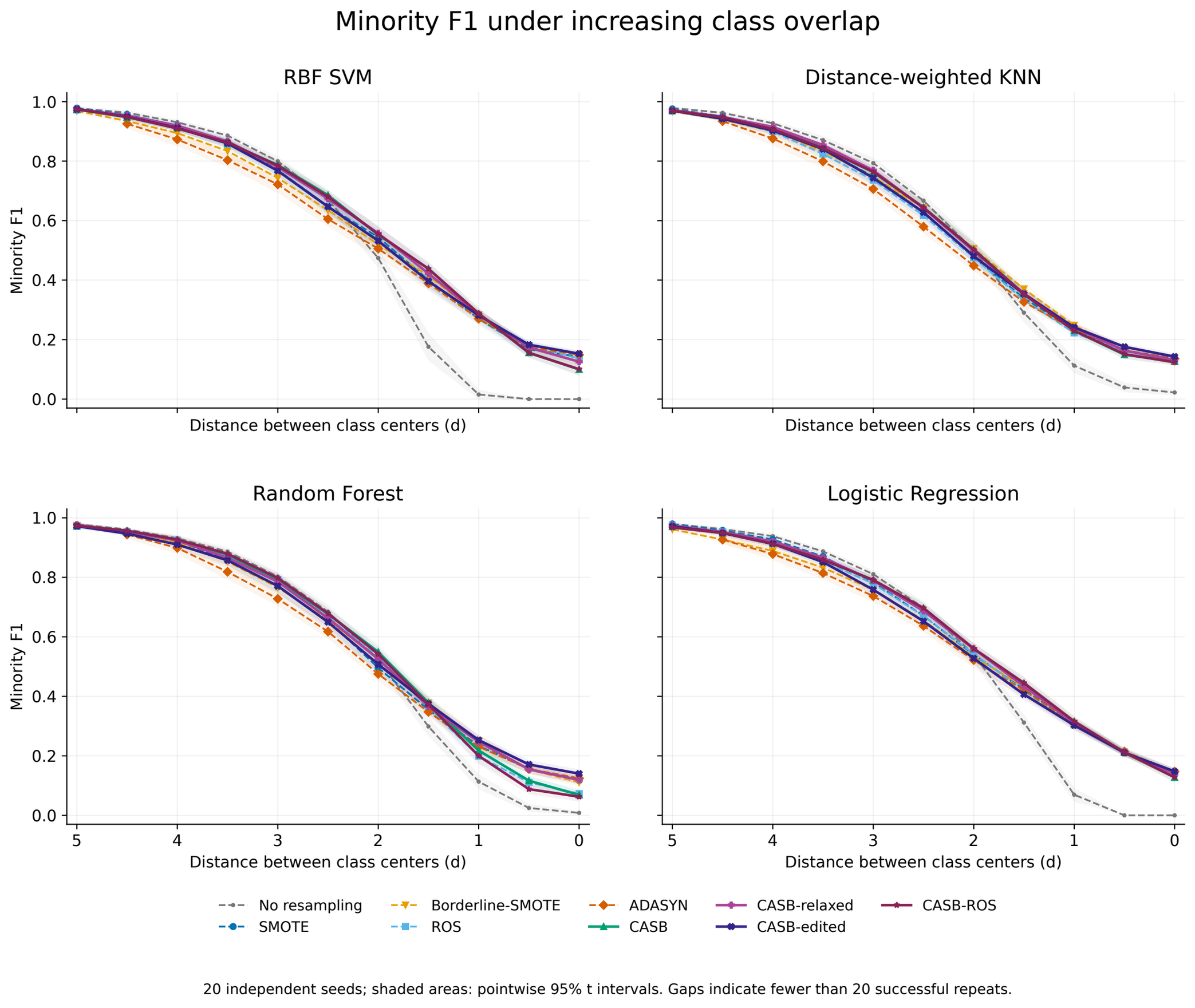
Minority F1 under increasing class overlap
RBF SVM
Distance-weighted KNN
Random Forest
Logistic Regression
Minority F1
1.0
0.8
0.6
0.4
0.2
0.0
5
4
3
2
1
0
Distance between class centers (d)
No resampling
SMOTE
Borderline-SMOTE
ROS
ADASYN
CASB
CASB-relaxed
CASB-edited
CASB-ROS
20 independent seeds; shaded areas: pointwise 95% t intervals. Gaps indicate fewer than 20 successful repeats.